

● HARD

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Circles

5 answers · Rationale-rich · Teach from doc

04

Q1**B****B. -14**

Choice B is correct. An equation of the form $x - h^2 + y - k^2 = r^2$, where h , k , and r are constants, represents a circle in the xy -plane with center h, k and radius r . Therefore, the circle represented by the given equation has center $-4, 19$ and radius 11. Since the center of the circle has an x -coordinate of -4 and the radius of the circle is 11, the least possible x -coordinate for any point on the circle is $-4 - 11$, or -15 . Similarly, the greatest possible x -coordinate for any point on the circle is $-4 + 11$, or 7. Therefore, if the point a, b lies on the circle, it must be true that $-15 \leq a \leq 7$. Of the given choices, only -14 satisfies this inequality.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2

A

A. $x^2 + y + 2^2 = 36$

Choice A is correct. The standard form of an equation of a circle in the xy -plane is $x - h^2 + y - k^2 = r^2$, where the coordinates of the center of the circle are h, k and the length of the radius of the circle is r . The equation of circle A, $x^2 + y - 2^2 = 36$, can be rewritten as $x - 0^2 + y - 2^2 = 6^2$. Therefore, the center of circle A is at 0, 2 and the length of the radius of circle A is 6. If circle A is shifted down 4 units, the y -coordinate of its center will decrease by 4; the radius of the circle and the x -coordinate of its center will not change. Therefore, the center of circle B is at 0, 2 - 4, or 0, - 2, and its radius is 6. Substituting 0 for h , -2 for k , and 6 for r in the equation $x - h^2 + y - k^2 = r^2$ yields $x - 0^2 + y - (-2)^2 = 6^2$, or $x^2 + y + 2^2 = 36$. Therefore, the equation $x^2 + y + 2^2 = 36$ represents circle B.

Choice B is incorrect. This equation represents a circle obtained by shifting circle A up, rather than down, 4 units.

Choice C is incorrect. This equation represents a circle obtained by shifting circle A right, rather than down, 4 units.

Choice D is incorrect. This equation represents a circle obtained by shifting circle A left, rather than down, 4 units.

Q3

5

The correct answer is 5. The standard form of an equation of a circle in the xy -plane is $x - h^2 + y - k^2 = r^2$, where h, k , and r are constants, the coordinates of the center of the circle are h, k , and the length of the radius of the circle is r . It's given that an equation of the circle is $x - 2^2 + y - 9^2 = r^2$. Therefore, the center of this circle is 2, 9. It's given that the endpoints of a diameter of the circle are 2, 4 and 2, 14. The length of the radius is the distance from the center of the circle to an endpoint of a diameter of the circle, which can be found using the distance formula, $\sqrt{x_1 - x_2^2 + y_1 - y_2^2}$. Substituting the center of the circle 2, 9 and one endpoint of the diameter 2, 4 in this formula gives a distance of $\sqrt{2 - 2^2 + 9 - 4^2}$, or $\sqrt{0^2 + 5^2}$, which is equivalent to 5. Since the distance from the center of the circle to an endpoint of a diameter is 5, the value of r is 5.

Q4

C

C. $x - 11^2 + y + 3^2 = 1$

Choice C is correct. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r units. It's given that circle A has the equation $(x - 7)^2 + (y + 3)^2 = 1$, which can be written as $(x - 7)^2 + (y - (-3))^2 = 1^2$. It follows that $h = 7$, $k = -3$, and $r = 1$. Therefore, the center of circle A is $(7, -3)$ and its radius is 1 unit. If circle A is translated 4 units to the right, the x -coordinate of the center will increase by 4, while the y -coordinate and the radius of the circle will remain unchanged. Translating the center of circle A to the right 4 units yields $(7 + 4, -3)$, or $(11, -3)$. Therefore, the center of circle B is $(11, -3)$. Substituting 11 for h , -3 for k , and 1 for r into the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 11)^2 + (y - (-3))^2 = 1$, or $(x - 11)^2 + (y + 3)^2 = 1$. Therefore, the equation $(x - 11)^2 + (y + 3)^2 = 1$ represents circle B.

Choice A is incorrect. This equation represents a circle obtained by shifting circle A down, rather than right, 4 units.

Choice B is incorrect. This equation represents a circle obtained by shifting circle A left, rather than right, 4 units.

Choice D is incorrect. This equation represents a circle obtained by shifting circle A up, rather than right, 4 units.

Q5

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Answer not provided in source data — see rationale below.

The correct answer is $\frac{1}{6}$. The circumference, C , of a circle is $C = 2\pi r$, where r is the length of the radius of the circle. For the given circle with a radius of 1, the circumference is $C = 2(\pi)(1)$, or $C = 2\pi$. To find what fraction of the circumference the length of arc $\overline{AB}$ is, divide the length of the arc by the circumference, which gives $\frac{\pi}{3} \div 2\pi$. This division can be represented by $\frac{\pi}{3} \cdot \frac{1}{2\pi} = \frac{1}{6}$. Note that $1/6$, $.1666$, $.1667$, 0.166 , and 0.167 are examples of ways to enter a correct answer.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q5 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

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